

# Margam Vishwateja

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## PROFESSIONAL SUMMARY

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Full Stack Developer with 2+ years of experience building production-grade IoT monitoring platforms, desktop applications, and mobile applications across the MERN stack. Delivered 10+ cross-platform desktop applications and 6+ mobile applications for radioactivity measurement and sensor data visualization. Deep expertise in real-time data pipelines (MQTT, WebSockets), signal processing with Python, modular plugin-based architecture, and RSA-based software licensing. Experienced across the full development lifecycle from embedded hardware integration to user-facing dashboards.

## TECHNICAL SKILLS

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**Frontend:** React.js, JavaScript (ES6+), HTML5, CSS3, Material UI (MUI), Redux, React Native, Next.js, Tailwind CSS, Recharts, Canvas.js, D3.js

**Backend:** Node.js, Express.js, REST API Development, WebSockets (ws), Cron Jobs, Child Processes

**Desktop & Mobile:** Electron.js (exe packaging, Webpack), React Native, FlutterFlow

**Databases:** MySQL, MongoDB, Redis, Firebase, Supabase, SQLite

**IoT & Real-Time:** MQTT, WebSockets, Serial Communication (PySerial), ESP32 OTA Firmware Updates

**Signal Processing:** Python, NumPy, SciPy, Gaussian Smoothing, FWHM, Spectral Calibration, ROI Analysis

**DevOps & Tools:** Git, CI/CD Pipelines, Docker (Basic), Linux, Webpack, Module Federation

**Security & Auth:** JWT Authentication, Middleware Authorization, RSA-based Licensing, Role-Based Access Control

**Other:** PDF/Excel Report Generation, Edge Functions, WordPress, FastAPI, CRM API Integration

## PROFESSIONAL EXPERIENCE

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### Full Stack Developer

Feb 2024 – Mar 2026

*Noki Technologies*

*Hyderabad, India*

- Designed and delivered 10+ Electron.js desktop applications and 6+ React Native mobile applications for scientific radioactivity measurement instruments, each tailored to a specific sensor type and shipped as standalone executables.
- Built real-time IoT data pipelines using MQTT and WebSockets to ingest live sensor telemetry from embedded devices and stream it to React dashboards with Recharts, Canvas.js, and D3.js visualizations.
- Implemented spectral energy calibration in Python (NumPy, SciPy) — converting raw detector channel data to energy values, computing Full Width at Half Maximum (FWHM), peak energy, and area under curve — invoked as a Node.js child process for seamless backend integration.
- Architected a modular skeleton framework that separated shared platform code (routing, WebSocket logic, Redux store, database queries, MQTT handlers) from application-specific modules, enabling new projects to launch with a ready-to-use template without touching the core codebase.
- Built a host application using Webpack Module Federation that unified multiple standalone device applications into a single dashboard, allowing operators to monitor, switch between, and configure multiple instruments without running separate applications.
- Implemented a dynamic multi-slot dashboard where users could assign live chart feeds from any connected application to any dashboard slot, enabling simultaneous monitoring of multiple devices in one view.
- Developed automated event-driven workflows using dual-queue architecture to execute sets of device instructions concurrently based on real-time IoT device status changes.
- Implemented RSA-based software licensing that binds each application to the installation machine with configurable time limits and per-instrument entitlements, preventing unauthorized usage.
- Built role-based access control (RBAC) across all desktop and mobile applications, restricting module access based on user privilege levels.
- Integrated serial communication via PySerial in Python child processes to receive raw sensor data over USB/serial, process it, and forward it to Node.js for dashboard delivery.

- Implemented automated background activity recording — applications automatically log sensor idle-state readings, trigger history saves, and enable individual or batch Excel/PDF report generation without user intervention.
- Built JWT-based authentication with Express.js middleware for API-level authorization, and used Redis for fast session storage across all platform applications.
- Developed ESP32 OTA (over-the-air) firmware update functionality within mobile applications, allowing remote firmware deployment to field devices.
- Implemented cron jobs for scheduled background tasks including automated data aggregation, history logging, and report generation pipelines.
- Wrote backend APIs using Express.js for device configuration management, history record creation, telemetry queries, and application-specific operations.
- Implemented CI/CD pipelines to automate builds and deployments, reducing manual release effort and deployment errors.

## FREELANCE EXPERIENCE

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### Travel Agency Web Application

React.js, Vite, Tailwind CSS, CRM API Integration

- Designed and developed a full travel agency website using React.js and Vite with Tailwind CSS; integrated backend APIs connecting with CRM systems for lead and booking management.

### Full Stack Web Development

Next.js, Python, FastAPI

- Developed full-stack web features using Next.js for frontend and FastAPI (Python) for backend REST API services.

## KEY PROJECTS

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### Unified Host Application – Multi-Device Dashboard

Electron.js, Webpack Module Federation, React.js, Redux

- Built a plugin-based host desktop application that consolidates multiple standalone device monitoring applications into one interface without merging codebases.
- Each application's frontend and backend was packaged as a self-contained Webpack plugin; new applications are added to the host simply by registering the plugin.
- Dashboard supports configurable slots — users assign live chart feeds from any application to any slot for simultaneous multi-device monitoring.

### Radioactivity Spectrum Analyzer

Python, NumPy, SciPy, Node.js, React.js, MQTT, Electron.js

- Developed desktop applications for gamma spectrum analysis — converting raw detector channel counts to energy values via user-defined calibration, identifying radioactive isotopes from energy peaks.
- Implemented Python-based spectral processing: Gaussian curve smoothing, FWHM computation, peak energy identification, and area-under-curve calculation using NumPy and SciPy.
- Integrated serial data acquisition via PySerial to receive raw counts directly from physical detectors over USB.

### Modular IoT Application Framework

Node.js, React.js, Redux, MQTT, WebSockets, MySQL

- Designed a reusable skeleton framework separating shared platform infrastructure from application-specific business logic, enabling 10+ applications to be built on a consistent, maintainable foundation.
- Skeleton includes pre-built modules for API routing, WebSocket server, MQTT handling, Redux store, and database query layers — application developers only write domain-specific code.

### React Native Mobile Suite – IoT Monitoring

React Native, Redux, MQTT, Node.js, SQLite, MySQL

- Built 6+ mobile applications for field use, mirroring the desktop platform's capabilities: live charts, MQTT data ingestion, local SQLite storage, PDF/Excel report generation, and RSA licensing.
- Implemented ESP32 OTA firmware update capability, logging with downloadable log files, and user authentication with role-based access control.

## EDUCATION

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### Kakatiya University

Bachelor of Technology – Computer Science and Engineering

Telangana, India

2019 – 2023